

# Ionomics

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## Protocol description

Trace elements in their ionic form mediate biochemical reactions in human cells by acting as enzyme cofactors or centers for stabilizing protein structures. Deficit or accumulation of these substances lead to cell toxicity and severe diseases in humans and therefore, intracellular trace ion concentrations (i.e. the "ionome") must be tightly controlled. Inductively coupled plasma mass spectrometry (ICP-MS) was used to determine and quantify the intracellular trace ion concentrations. In ionomics assays HEK cells overexpressing a particular doxycycline-inducible SLC transporter were used to quantify the change of inorganic ions by ICP-MS upon cell lysis. ICP-MS based ionomics is rather a low throughput assay, as a significant volume of sample is required.

## Materials

|                              |                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                    |
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| <b>Biological materials:</b> | <p>Human embryonic kidney 293 with doxycycline-inducible expression of human SLCc from CeMM:</p> <p>Control-Cells:</p> <ul style="list-style-type: none"><li>▪ HEK293 JumpIn T-Rex cells WT (+ dox and - dox)</li><li>▪ CE026N-V SLC12A6 (+ dox and - dox) as unrelated SLC cell line</li><li>▪ CE02FM-C SLC39A14 (+ dox and - dox)</li><li>▪ CE02GZ-7 SLC40A1 (+ dox and - dox)</li></ul> <p>Cells of interest:</p> <ul style="list-style-type: none"><li>▪ CE02BA-5 SLC30A8-WTOE (+ dox and - dox)</li><li>▪ CE07CM-M SLC30A8_D110N_D224N-WTOE (+dox and - dox)</li></ul>                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                        |
| <b>Reagents:</b>             | <ul style="list-style-type: none"><li>▪ DMEM supplemented 500ml (Gibco Cat:31966) with 10% fetal bovine serum (FBS, Gibco Cat:10500-064, Lot:2534346H), 100µg/mL streptomycin and 100 IU/mL penicillin (Gibco, Cat:15140-122) with 2 mg/mL G418 (Gibco, Cat:10131-027) and 5µg/mL Blasticidin S HCl (Gibco Cat: A11139-03), +/- doxycycline (10ng/mL, Sigma).</li><li>▪ For JumpIn cells instead of G418 use 2ml Hygromycin B (Invitrogen Cat:10687010), 200ug/ml Medium.</li><li>▪ HBSS-uptake buffer (Gibco Cat:14025): 12 mM HEPES (Gibco Cat:15630), 0.8 mM L-Ascorbic acid (Sigma Cat:795437), and 0.4% bovine-serum albumin (Sigma Cat: A7030)</li><li>▪ Metal Ion mix in uptake buffer: 4µM (NH<sub>4</sub>)<sub>5</sub>[Fe(C<sub>6</sub>H<sub>4</sub>O<sub>7</sub>)<sub>2</sub>] (ferric-ammonium citrate), 6µM ZnCl<sub>2</sub> (Fluka Cat:96468), 3µM CuCl<sub>2</sub>(Sigma Cat:C3279), 3µM (Sigma MnCl<sub>2</sub> 7xH<sub>2</sub>O), 3µM CoCl<sub>2</sub> (Sigma Cat:60818) and 3µM NiCl<sub>2</sub>x7H<sub>2</sub>O (SIGMA Cat: N6136).</li><li>▪ Washing buffer: isotonic Tris-choline-chloride (Sigma Cat: C7527), D-Glucose (Sigma Cat: G7021)</li><li>▪ Lysis buffer: Tris-choline-chloride with Triton X-100 (Sigma Cat: 93443)</li><li>▪ Measure of Protein concentration, Pierce BCA Protein Assa Kit (Thermo scientific Cat:23225)</li></ul> |
| <b>Equipment:</b>            | <ul style="list-style-type: none"><li>▪ ICP-MS, Element 2, Thermo Fisher Scientific</li></ul>                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                      |

## Reagents setup

Growth Medium SLC-cells:

500ml DMEM supplemented with 10% fetal bovine serum, 100 µg/mL streptomycin and 100 IU/mL penicillin, with 2 mg/mL G418 and 5µg/mL Blastidicin S HCl.

Growth Medium JumpIn-cells:

500ml

DMEM supplemented with 10% fetal bovine serum, 100 µg/mL streptomycin and 100 IU/mL penicillin, with 2ml Hygromycin B and 5 µg/mL Blastidicin S HCl.

Assay plating medium:

500ml

DMEM supplemented with 10% fetal bovine serum, 100 µg/mL streptomycin and 100 IU/mL penicillin.

6-Well Plates:

Poly-D-Lysine, #657940 from Greiner Bio-One

3xDoxycycline induction Medium:

Doxycycline

in dH<sub>2</sub>O (conc 1mg/ml) dissolved 60ul in 9940ul dH<sub>2</sub>O (6ug/ml), used the stock to make a 3x Dox induction Medium: 125ul stock solution in 24875ul Assay Medium (30ng/ml) end conc. 10ng/ml, sterilize through a 0.22-µm filter.

Washing Medium: 500ml DMEM supplemented without FBS

Prepare BSA 3%:

Weigh out 1500 mg of BSA in a 50ml Falcon tube and fill up to 50 ml mark with (12mM Hepes + HBSS).

Prepare 100mM Ascorbic acid:

Weigh out 616.4 mg of Ascorbic acid in a 50ml Falcon tube. Dissolve in 35ml of dH<sub>2</sub>O. This is the 100mM Stock solution

Uptake Buffer (HBSS/12mM HEPES/0.8mM Ascorbic acid/0.4% BSA):

In HBSS 177800ul add 2520ul Hepes (1M) add AA 100mM 1512ul add 28000ul BSA (3%), sterilize through a 0.22-µm filter.

Prepare the 1mM intermediate dilution of each metal in dH<sub>2</sub>O from the frozen 10mM metal stock solution:

200ul Metal

stock solution (10mM) add 1800ul dH<sub>2</sub>O.

Prepare the 200ml Flask containing the different compositions of uptake buffer and metal mix:

Uptake buffer 15600ul add FAC 1mM 800ul, ZnCl<sub>2</sub> 1mM 1200ul all other 4xMetal (CuCl<sub>2</sub>, MnCl<sub>2</sub>, CoCl and NiCl<sub>2</sub>) each 600ul of 1mM solution and sterilize filter 0.22-µm.

Preparation of choline chloride solution:

2L of a 141mM solution of choline chloride in dH<sub>2</sub>O. Weight out 39.372g of choline chloride and fill up to 2L with dH<sub>2</sub>O and sterilize through a 0.22-µm filter.

Preparation of Glucose 1M:

90.08g Glucose in 500ml dH<sub>2</sub>O sterilize filtered 0.22-µm store in the fridge 4°C

Sodium free sample wash buffer:

Prepare 1000ml of sample wash buffer in a 1l glass bottle by mixing the choline chloride solution (141mM) 970ml, add Glucose (1M) 5ml, add Tris-HCl pH 7.4 (1M) 25ml.

Sodium free lysis buffer:

Prepare 400ml of sodium free sample lysis buffer by mixing the choline chloride 386ml solution with Tris-HCl 10ml in a sterile 500ml bottle.

Add a stirring bar to the bottle and slowly add 4ml of 100% Triton X-100 and let the viscous liquid completely dissolve for an hour with constant stirring. Then put the bottle on ice to be cold for the cell lysis step.

To analyze the amounts of intracellular ions (<sup>23</sup>Na, <sup>24</sup>Mg, <sup>44</sup>Ca <sup>31</sup>P, <sup>32</sup>S, <sup>55</sup>Mn, <sup>56</sup>Fe, <sup>63</sup>Cu, <sup>52</sup>Cr, <sup>59</sup>Co, <sup>57</sup>Mo <sup>60</sup>Ni <sup>66</sup>Zn, <sup>51</sup>V and <sup>39</sup>K), the lysate was first diluted with nitric acid (1.9% m/V, Ultra-Quality from Roth or equivalent) and then the internal standard Thulium was added (1 µg/l, TraceCERT from Sigma or equivalent). High resolution ICP-MS analysis was performed using ELEMENT2 ICP-MS (Thermo Scientific) and analyzed using the ELEMENT2 ICP-MS software (Thermo Scientific). The final ion concentrations were normalized to either cell lysate protein concentration (BCA, Thermo Fischer Scientific) or amount of <sup>31</sup>P.

## Procedure

### Part A Influx protocol

#### ▪ Part A of protocol / day 1

1. step 1 Plate cells (650´000 cells/well) in 6 well plates Poly-D-Lysin in Assay Medium

#### ▪ Part A of protocol / day 2

2. Induce transgene expression by addition of 10ng/ml Doxycycline in medium/Pen/Strep without FBS (20h)  
Untreated (n=1) or  
Dox 1µg/ml (used broadly in ionomics) or 10 ng/ml, used for selected assays (n=5)

#### ▪ Part A of protocol / day 3

3. Aspirate off Assay medium containing dox and replace by an uptake buffer containing a metal mix.
4. Metal mix used for spiking (final conc., 1x): Fe (4µM), Zn (6µM), Cu (3µM), Mn (3µM), Co (3µM), Ni (3µM) in Uptake buffer. Uptake buffer used for spiking protocol: HBSS + 12mM Hepes + 0.4% BSA + 0.8 mM Ascorbic acid.
5. Incubate for 3h at 37°C before cells are washed and lysed on ice.
6. Wash cells 3x with 3ml wash buffer.
7. Lyse cells in 700µl Lysing buffer, incubate 30min on ice .

8. Pipette up and down all time on ice and collected the complete lysate in an 1.5ml Eppendorf.
9. The lysates are vortexed in the tube and incubated on ice for another 15min. Then they are centrifuged at 20'000xg for 20minutes in the cooled centrifuge and the supernatant is transferred to a fresh 2ml Eppendorf tube.
10. Measure Protein use (3x 10µl Lys) Pierce BCA Protein Assay Kit, the rest store in -80C until send to St Gallen. Ship samples on dry ice to St Gallen/Anna-Lena Steck and keep at -20/-80°C until analysis for determine ion conc. by ICP-MS.

## Part B Efflux protocol

### ▪ Part B of protocol / day 1

1. step 1 Plate cells (1 ' 1000 ' 000 cells/well) in 6 well plates Poly-D-Lysin in Assay Medium/Ascorbic acid 0.8mM + metal loading 48uM Zn or 96uM Zn or 12uM Fe

### ▪ Part B of protocol / day 2

2. Remove medium with loading Metal (wash 2x cells with 2ml of wash Medium) and replace with 2ml D-MEM w/o FCS + Pen/Strep (serum free medium)  
+/- Dox and for SLC40A1 additional VIT-2763

### ▪ Part B of protocol / day 3

3. collect supernatant (2ml)
4. Wash cells 3x with 3ml wash buffer
5. Lyse cells in 700µl Lysing buffer, incubate 30min on ice .
6. Pipette up and down all time on ice and collected the complete lysate in an 1.5ml Eppendorf.
7. The lysates are vortexed in the tube and incubated on ice for another 15min. Then they are centrifuged at 20'000xg for 20minutes in the cooled centrifuge and the supernatant is transferred to a fresh 2ml Eppendorf tube.
8. Measure Protein use (3x 10µl Lys) Pierce BCA Protein Assay Kit, the rest store in -80C until send to St Gallen. Ship samples on dry ice to St Gallen/Anna-Lena Steck and keep at -20/-80°C until analysis for determine ion conc. by ICP-MS.

## Additional notes

## Data availability References

Dvorak V et al., Front. Pharmacol. 2021  
Polesel M et al, Biometals, 2022